

JULIANA SANTACRUZ RESTREPO

Junior Data Scientist | Machine Learning | Python | Analytics | Cloud (GCP)

Pasto, Nariño - Colombia | Spanish Citizenship (EU Passport)

julianasantarestrepo@gmail.com | +57 321 844 5021

Open to travel or relocation

PROFESSIONAL PROFILE

Junior Data Scientist with a background in Systems Engineering and a Master's in Data Science. Experienced in designing end-to-end machine learning solutions, including data preprocessing, feature engineering, and predictive modeling using large-scale datasets.

Hands-on experience working with distributed data processing (PySpark) and developing anomaly detection models for cybersecurity applications in connected vehicles. Proficient in Python, applied statistics, and cloud-based analytics in Google Cloud Platform (GCP).

Currently expanding my expertise in LLMs (LangChain, GPT APIs) and prompt engineering, along with NoSQL databases (MongoDB) to enhance my capabilities in applied AI systems.

PROFESSIONAL EXPERIENCE

Data Analyst – Data Rookie

Davinci Technologies | September 2023 - July 2024

- Performed data cleaning, preprocessing, and exploratory data analysis (EDA) on structured datasets to extract actionable insights.
- Developed machine learning models for predictive analysis using Python and Scikit-learn.
- Applied feature engineering techniques to improve model performance and interpretability.
- Supported data-driven decision-making by generating reports and visualizations.
- Worked with Google Cloud Platform (BigQuery) for querying and analyzing large datasets.
- Built analytical visualizations to communicate findings to both technical and non-technical stakeholders.

Development Intern

JHWEB S.A.S | January 2023 - May 2023

- Supported the development of technological solutions, strengthening programming foundations and computational logic.
- Managed relational databases (MySQL) to support various applications.

PROJECTS

Cyberattack Risk Analysis in Connected Vehicles (Final Master Project)

- Designed an end-to-end data pipeline to detect cyberattacks (DoS, spoofing, fuzzing) using CAN bus data.
- Processed over 17M records using PySpark for distributed data analysis.
- Performed data cleaning, feature engineering, and exploratory data analysis (EDA) to identify anomalous patterns in CAN bus signals.
- Developed supervised and unsupervised models (Random Forest, XGBoost, Autoencoders) for anomaly detection.
- Applied feature engineering and class imbalance techniques to improve model performance.

Decision Support System for COVID-19 University Reopening (Undergraduate Thesis)

- Developed a decision-support application to evaluate safe return-to-campus scenarios based on health and capacity variables.
- Modeled decision criteria including symptoms, social distancing, and classroom occupancy constraints.
- Designed logical rules to support risk assessment and operational decision-making.
- Built a functional prototype using FlutterFlow to simulate real-world usage.

EDUCATION

Master’s Degree in Data Science

CEUPE - Spain | 2024 - 2025

- Focus on Machine Learning, Big Data, and Predictive Analytics
- Final Project: Cyberattack Risk Analysis in Connected Vehicles (Grade: 100/100)

Bachelor’s Degree in Systems Engineering

Universidad Mariana - Colombia | 2019 - 2023

- Graduated with Honors for Outstanding Final Project

TECHNICAL SKILLS

- **Programming:** Python, Java
- **Data Science:** Machine Learning, Anomaly Detection, Predictive Modeling, EDA
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PySpark, Spark MLlib
- **Big Data & Cloud:** Google Cloud Platform (BigQuery), Apache Spark, Distributed Data Processing
- **Databases:** SQL (MySQL, PostgreSQL), Basic knowledge of NoSQL (MongoDB), Firebase.
- **AI & Emerging Tech:** Foundations in LLM applications (LangChain, GPT APIs)
- **Visualization:** Power BI, Looker Studio, Plotly, Matplotlib, Seaborn
- **Tools:** Git, GitHub, GitLab, Jupyter Notebook

CERTIFICATIONS & CONTINUOUS LEARNING

- Responsible Prompting: Maximizing AI in Business – Microsoft & Founderz
- Google Cloud Big Data and Machine Learning Fundamentals.
- Building Batch Data Pipelines on Google Cloud.
- Modernizing Data Lakes and Data Warehouses with Google Cloud.
- Smart Analytics, Machine Learning, and AI on Google Cloud.
- NDG Linux Essentials – Cisco Networking Academy.

ACADEMIC ACHIEVEMENTS & ACTIVITIES

- Speaker at national and international conferences on science and engineering (2021 - 2023).
- Active participation in research groups and academic events.

LANGUAGES

- Spanish: Native.
- English: B2 (Upper-intermediate).